12. Design a C Program to Simulate the Concept of Dining-Philosophers Problem

Code - Part 1 (Lines 1–35)

```c
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <pthread.h>
4  #include <unistd.h>
5
6  #define NUM_PHILOSOPHERS 5
7
8  pthread_mutex_t chopsticks[NUM_PHILOSOPHERS];
9
10 void* philosopherLifeCycle(void* arg)
11 {
12     int id = *((int*)arg);
13
14     int left = id;
15     int right = (id + 1) % NUM_PHILOSOPHERS;
16
17     while (1)
18     {
19         // Thinking
20         printf("Philosopher %d is thinking...\n", id);
21         sleep(1);
22
23         // Deadlock prevention:
24         if (id % 2 == 0)
25         {
26             pthread_mutex_lock(&chopsticks[left]);
27             pthread_mutex_lock(&chopsticks[right]);
28         }
29         else
30         {
31             pthread_mutex_lock(&chopsticks[right]);
32             pthread_mutex_lock(&chopsticks[left]);
33         }
34
35         // Eating
```

Code - Part 2 (Lines 36–70)

```
36          printf("Philosopher %d is eating...\n", id);
37          sleep(rand() % 3 + 1);
38
39          // Put down chopsticks
40          pthread_mutex_unlock(&chopsticks[left]);
41          pthread_mutex_unlock(&chopsticks[right]);
42      }
43
44      return NULL;
45  }
46
47  int main()
48  {
49      pthread_t philosophers[NUM_PHILOSOPHERS];
50      int philosopher_ids[NUM_PHILOSOPHERS];
51
52      // Initialize mutexes
53      for (int i = 0; i < NUM_PHILOSOPHERS; i++)
54      {
55          pthread_mutex_init(&chopsticks[i], NULL);
56      }
57
58      // Create threads
59      for (int i = 0; i < NUM_PHILOSOPHERS; i++)
60      {
61          philosopher_ids[i] = i;
62          pthread_create(&philosophers[i], NULL, philosopherLifeCycle,
63                         &philosopher_ids[i]);
64      }
65
66      // Join threads (runs forever)
67      for (int i = 0; i < NUM_PHILOSOPHERS; i++)
68      {
69          pthread_join(philosophers[i], NULL);
70      }
```

Output

```
Philosopher 0 is thinking...
Philosopher 1 is thinking...
Philosopher 3 is thinking...
Philosopher 2 is thinking...
Philosopher 4 is thinking...
Philosopher 0 is eating...
Philosopher 3 is eating...
Philosopher 3 is thinking...
Philosopher 0 is thinking...
Philosopher 1 is eating...
Philosopher 4 is eating...
Philosopher 1 is thinking...
Philosopher 2 is eating...
Philosopher 4 is thinking...
Philosopher 0 is eating...
Philosopher 2 is thinking...
Philosopher 3 is eating...
Philosopher 0 is thinking...
Philosopher 1 is eating...
Philosopher 1 is thinking...
Philosopher 0 is eating...
Philosopher 3 is thinking...
Philosopher 2 is eating...
Philosopher 0 is thinking...
Philosopher 4 is eating...
Philosopher 2 is thinking...
Philosopher 1 is eating...
Philosopher 4 is thinking...
Philosopher 3 is eating...
Philosopher 1 is thinking...
Philosopher 0 is eating...
Philosopher 0 is thinking...
Philosopher 3 is thinking...
Philosopher 2 is eating...
Philosopher 4 is eating...
Philosopher 4 is thinking...
```